# Weishaupt WTC with WEM – who talks to whom

Gas condensing boiler WTC-GW (B series) with built-in WEM system unit · ESP bridge from this repository

**Internet – Weishaupt cloud**

**Device server**
xmpp.wemportal.de · Port 5222
XMPP over TLS (STARTTLS)
only WEMs with a device certificate

internal at Weishaupt

**WEM portal (website)**
wemportal.com
display, remote control, data logging, fault messages

HTTPS

**Browser / Weishaupt app**
from anywhere

encrypted (TLS 1.2)

**Router – home network ↔ internet**

**Home network – unencrypted**
the WEM opens the connection, keep-alive about once a minute

**WEM Modbus gateway**
Modbus TCP for automation, polls the WEM cyclically

LAN · JSON

**Phone app in Docker**
like the app on the board, via MQTT · folder app/

**MQTT broker**
optional: InfluxDB, Grafana, Home Assistant

**one LAN socket,** one cable

HTTP / JSON via Wi-Fi + router – on demand only

Wi-Fi · MQTT · HTTP (IP, same as LAN)

**BOILER UNIT (Weishaupt WTC)**

**WEM-SG system unit (node 1)**
= display / control panel of the unit
- **heating curve** → flow setpoint
- heating circuit / hot water, schedules
- fault memory (8 bytes, CAN segmented)
- data logger for the portal
- JSON interface (ESP, gateway)
- SD card, network socket

system master

**BOILER – burner control**
WEM-FA-G (node 2)
- applies the flow setpoint
- flame, fan, gas valve
- boiler temp., flue gas, flow rate
- operating hours, burner starts
- own limits (flue gas, anti-cycling, pump), flame supervision
- WEM missing: fault F 91

WEM writes to the boiler every ~5 s: flow setpoint, DHW setpoint, demand

WEM polls the boiler ~3×/s; boiler writes readings into the WEM

2 wires

**CAN bus 50 kbit/s (CANopen SDO/PDO) – unencrypted**
every node hears everything · one SDO reply carries max. 4 bytes

**ESP bridge (WeAct CAN485)**
- listens to everything
- reads single values (SDO)
- switches via CAN at the WEM, verifies the result on the bus
- phone app built in (browser)

fallback: switching via WEM JSON

**JSON = remote CAN access over LAN (CanApiJson)**
A request names a CAN object, e.g. module 07 · 2533/02 = node 2.
The WEM translates, reads or writes on the bus and answers as JSON.
It is translator and gatekeeper: only permitted objects, nothing on the boiler is writable. Its own objects it answers directly.

**Room sensor / room unit**
WEM-RF, WEM-RG · in the living area on the same CAN bus

optional · **from this repository** · CAN bus · through the router
encrypted · home network, unencrypted · one LAN socket on the WEM

Sources: CAN capture, network capture, connection test to the device server, Weishaupt manuals WTC-GW 15…32-B (83307201) and WEM Modbus gateway (83325901).